Ailoni Sahasransh Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Sri Chaitanya Junior Kalasala, DD Colony, Hyderabad	Class XII (Telangana State Board of Intermediate Education)	2024	97.00%
Sri Aurobindo International School, Vidyanagar, Hyderabad	Class X (Indian Certificate of Secondary Education)	2022	97.40%

SCHOLASTIC AND TECHNICAL ACHIEVEMENTS

- Qualified for **INChO 2023-24** and **INAO 2023-24** Olympiads conducted by **HBCSE**.
- **National Top 1** Percentile in **NSEP 2023-24**, **NSEC 2023-24**, and **NSEA 2023-24** conducted by **IAPT**.
- Secured Rank **5** in **IICPC Quantfest 2025 Finals**, after securing Rank **77** in **Regionals** and Rank **225** in **Prelims**.
- Secured Rank **81** in **IICPC Codefest 2026 Regionals**, after securing Rank **560** in **Prelims**.
- Finished **11th** in **Optiver Trade-A-Thon** held on 23rd March 2026 among 400+ participants.
- All India Rank **122** in **ICPC 2025 Prelims**, **4th** in IIT Hyderabad.
- Codeforces: Handle-atigdng; Peak Rating **1741** (Expert)
- CodeChef: Handle-atigdng; Peak Rating **1617** (3-stars)
- Part of **2nd placed** team in **Jane Street Estimathon** and **DaVinci Battle Royale** during **IICPC Quantfest 2025 Finals**.

SKILLS

- **Languages**: C++, C, Bash, RISC-V assembly, Python, HTML.
- **Tools and technologies**: LaTeX, PostgreSQL, Linux, Git.

PROJECTS

Hybrid Core Scheduling for Performance on Linux(Group Project, 2 members) | GitHub Link

- Developed an **adaptive thread-level scheduler** for Intel hybrid CPUs, balancing workloads across **P-cores** and **E-cores**.
- Built a **real-time monitor** using /proc and turbostat to track CPU usage, IPC, frequency, power, and temperature.
- Designed a **swap-based promotion engine** via sched_setaffinity() with **persistence counters** to reduce migrations.
- Implemented **thermal state-machine** and admission control to prevent P-core saturation and improve stability.
- Achieved **~15%** lower execution time at **~7.1%** higher power, with only **~1%** throughput loss from migrations.

Enhancing xv6 Operating System | GitHub Link

- Added custom system calls for process management, parent-child information, and per-process syscall statistics.
- Replaced round-robin scheduler with a **4-level MLFQ scheduler** featuring priority boosting and starvation prevention.
- Implemented **Clock page replacement** with a global frame table and per-process memory accounting.
- Extended scheduler, VM, traps, and process management; validated using comprehensive user-space test programs.

Floating-Point Performance Optimization | GitHub Link

- Converted integer-heavy C programs to **floating-point implementations** to better utilize modern CPU execution units.
- Applied loop unrolling, blocking, and multiple accumulators to improve **instruction-level parallelism**.
- Achieved up to **21% speedup** for array maximum; observed moderate gains for matrix multiplication and dot product.
- Evaluated cache hierarchy, vectorization, and compiler optimizations using **gem5** and Intel hardware.

Statistical Data Analysis | Link to reports

- Processed real-world datasets to generate **frequency distributions** and statistical visualizations.
- Applied **sampling distributions**, **point estimation**, and **confidence interval estimation** on multiple datasets.
- Performed left-, right-, and two-tailed **hypothesis tests** and analyzed **2×2 contingency tables**.
- Developed **linear regression** models to quantify variable relationships and interpret model parameters.

RELEVANT COURSES

- **Mathematics:** Discrete Mathematics, Probability and Random Variables, Number Theory, Applied Statistics, Introduction to Statistics, Linear Algebra, Calculus, Series of Functions, Transform Techniques.

- **Computer Science:** Data Structures, Algorithms, Software Development Fundamentals, Theory of Computation, Computer Architecture, Operating Systems, Database Management Systems, Compilers, Digital Circuits, Artificial Intelligence, Introduction to Computing, Introduction to Programming.

POSITIONS OF RESPONSIBILITY

Senior Core Member at Infero: The Algorithmic and Logical Reasoning club at IIT Hyderabad.

- Collaborating with heads to organize competitive programming contests, and community sessions.
- Mentoring cores through problem-solving sessions and contest preparations.

Core Member at Infero: The Algorithmic and Logical Reasoning club at IIT Hyderabad.

- Mentored students in Competitive Programming through training sessions and contests.
- Contributed to problem setting for **Elan & NVision CodeClash 2026**.

Co-ordinator at FCC Quant: The Quant Domain in the Finance club at IIT Hyderabad.

- Helped coordinate club operations and event execution as part of the Quant team.

Teaching Assistant for CS1010: Discrete Mathematics for Computer Science

- Managed academic support and doubt-clarification for 150+ students while evaluating examinations.

EXTRACURRICULAR

- Represented IIT Hyderabad in **Chess** at the **Inter IIT Sports Meet 2025**.

- **NSO Cricket** player at IIT Hyderabad.